Lab # 03

Bubble, Selection, Insertion Sort

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Data Structures and Algorithms Lab

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Task 01: Implementation of Bubble Sort in Arrays:

sort_T_03.cpp | bubble_sort_T_01.cpp | powershell

lab_03_sorting_algorithms > bubble_sort_T_01.cpp > bubbleSort(int [], int)

You, 2 hours ago | 1 author (You)

```cpp
#include <iostream>
using namespace std;
void bubbleSort(int arr[], int size)
{
    for (int i = 0; i < size - 1; i++)
    {
        for (int j = i + 1; j < size; j++)
        {
            int temp;
            if (arr[i] > arr[j])
            {
                temp = arr[i];
                arr[i] = arr[j];
                arr[j] = temp;
            }
        }
    }
    for (int i = 0; i < size; i++)
    {
        cout << arr[i] << " ";
    }
}    You, 3 hours ago • bubble sort implemented
int main()
{
    int arr[] = {3, 5, 1, 2, 6, 4};
    int size = sizeof(arr) / sizeof(arr[0]);
    bubbleSort(arr, size);
}
```

```
 2 files changed, 34 insertions(+)
 create mode 100644 lab_03_sorting_algorithms/insertion_sort_T_03.cpp
PS D:\4th Semester\DSA\lab_03_sorting_algorithms> git push -u origin mai
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 98.60 KiB | 7.04 MiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/mohsinkhan117/CPP_DSA.git
   1ea16d2..ff1aa26  main -> main
branch 'main' set up to track 'origin/main'.
PS D:\4th Semester\DSA\lab_03_sorting_algorithms> g++ .\bubble_sort_T_01
PS D:\4th Semester\DSA\lab_03_sorting_algorithms> .\a.exe
1 2 3 4 5 6
PS D:\4th Semester\DSA\lab_03_sorting_algorithms>
```

Task 02: Selection sort implementation:

selection_sort_T_02.cpp | insertion_sort_T_03.cpp | bubble_sort_T_0' | powershell

lab_03_sorting_algorithms > selection_sort_T_02.cpp > main()

```cpp
#include <iostream>
using namespace std;

void selectionSort(int arr[], int size)
{
    for (int i = 0; i < size - 1; i++)
    {
        int min = i;

        for (int j = i + 1; j < size; j++)
        {
            if (arr[j] < arr[min])
            {
                min = j;
            }
        }

        if (min != i)
        {
            swap(arr[i], arr[min]);
        }
    }
}

int main()
{
    int arr[] = {3, 5, 1, 2, 6, 4};
    int size = sizeof(arr) / sizeof(arr[0]);

    selectionSort(arr, size);

    for (int i = 0; i < size; i++)
    {
        cout << arr[i] << " ";
    }

    return 0;
}
```

```
PS D:\4th Semester\DSA\lab_03_sorting_algorithms> g++ .\selection_sort
_02.cpp
PS D:\4th Semester\DSA\lab_03_sorting_algorithms> .\a.exe
1 2 3 4 5 6
PS D:\4th Semester\DSA\lab_03_sorting_algorithms>
```

Task 03: Implementation of Insertion Sort:

selection_sort_T_02.cpp | insertion_sort_T_03.cpp | bubble_sort_T_0

lab_03_sorting_algorithms > insertion_sort_T_03.cpp > main()

```cpp

#include <iostream>
using namespace std;

void insertionSort(int arr[], int n)
{
    int i, key, j;

    // Start from the second element (i=1)
    for (i = 1; i < n; i++)
    {
        key = arr[i]; // Element to be inserted
        j = i - 1;    // Start comparing with the last element

        while (j >= 0 && arr[j] > key)
        {
            arr[j + 1] = arr[j];
            j = j - 1;
        }
        arr[j + 1] = key; // Insert the key into its correct
    }
    for (int i = 0; i <= n; i++)
    {
        cout << arr[i] << " ";
    }
}
int main()
{
    int arr[] = {12, 11, 13, 5, 6};
    int n = sizeof(arr) / sizeof(arr[0]);
    insertionSort(arr, n);

    return 0;
}
```

powershell

```
PS D:\4th Semester\DSA\lab_03_sorting_algorithms> g++ .\insertion_sort_T_03.cpp
PS D:\4th Semester\DSA\lab_03_sorting_algorithms> .\a.exe
5 6 11 12 13 0
PS D:\4th Semester\DSA\lab_03_sorting_algorithms>
```

Comparison :

Feature	Bubble	Selection	Insertion
Best Case	$O(n)$	$O(n^2)$	$O(n)$
Average Case	$O(n^2)$	$O(n^2)$	$O(n^2)$
Swaps/Moves	Many	Few swaps	Fewer operations
Stable	Yes	No	Yes
Adaptive	Yes	No	Yes
Practical Performance	Poor	Fair	Good

RUBRICS (LABS/ PROJECTS):

Criteria & Point Assigned	**Outstanding 4**	**Acceptable 3**	**Considerable 2**	**Below Expectations 1**	**Score**
Valuing related to work, including responsibility, and professionalism.	Demonstrates **excellent** responsibility and professionalism in task managment. Shows **strong commitment to** quality work, meeting the deadline, and comprehensive, well-organized **effective documentation** following all guidelines with exemplary attention to detail.	Demonstrates **some** responsibility and professionalism. Displays **moderate commitment to** quality work, meets the deadline and shows **sufficient thoroughness** in documentation, adhering to guidelines, with some room for enhancement.	Demonstrates **limited effort** in responsibility and professionalism. Misses the deadline with **notable inconsistency** in quality work and shows **insufficient** thoroughness or clarity in documentation, necessitating improvements.	Demonstrates a **lack** of responsibility and professionalism. Misses the deadline and **lacks** quality work. Documentation is incomplete, disorganized, or unclear.	
Ability to apply techniques, methods, and procedures effectively.	Executes lab tasks **with exceptional skill** to design/develop solution and **shows proficiency** in lab techniques and procedures.	Executes lab tasks **with adequate skill** to design/develop solution and **shows proficiency** in lab techniques and procedures.	Executes lab tasks **with minimal skill** to design/develop solution and **shows modeate proficiency** in lab techniques and procedures.	Executes lab tasks **with limited skill** to design/develop solution and **lacks proficiency** in lab techniques and procedures.	

Demonstrate understanding and relevant Knowledge.	Demonstrates **exceptional** understanding of concepts with **clear explanation** of investigation and analysis of the objectives and outcomes.	Demonstrates a **good** understanding of concepts with **some explanations** of investigation and analysis that align with the lab's objectives and outcomes.	Demonstrates a **basic** understanding of key concepts but **lacks clarity or depth in explaining** the investigation and analysis, with some inconsistencies.	Demonstrates **limited** understanding of concepts with **unclear explanations** of investigation and analysis that do not align with the lab’s objectives and outcomes.	